5T+1 CFET SRAM with Dual-Bitline-per-Cell-Height Enabling Differential Readout and 12.5% Area Reduction

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Abstract: This work presents the 5T+1 CFET SRAM, a 5T-footprint bitcell exploiting vertical n/p stacking to integrate a transmission-gate access device without area penalty. A dual-bitline-per-cell-height architecture is proposed to enable differential sensing using the bitline of an unselected cell as a reference. The bitcell integrates a split-V_{DD} in-bitcell write-assist to improve writability without impacting bitcell footprint. DTCO evaluation shows 12.5% smaller bitcell than 6T CFET, 7–15% faster write, and 21% standby leakage reduction, while maintaining comparable RSNM and >150 mV five-percentile HSNM for unselected cells during write under split-V_{DD}.

Introduction: Continued scaling of SRAM bitcells in the nanosheet and CFET era relies on device-architecture innovations, such as forksheet and CFET integration **(Fig. 1)**. While prior CFET-based SRAM studies demonstrate aggressive scaling of conventional 6T bitcells [1-3], further area reduction is constrained by incomplete utilization of the CFET stack, as unused devices must be selectively removed to realize a 6T topology **(Fig. 2(a))** [1]. This inefficiency arises from the inherent 4/2 p/n (or n/p) device imbalance of the SRAM bitcell, requiring four n/p gate pairs in a CFET implementation and leaving two devices unused (**Fig. 2(a)**). As a result, the intrinsic vertical density advantage of CFET is not fully translated into bitcell area scaling for 6T SRAM.

5T SRAM bitcells **(Fig. 2(b))** have been explored as a density-driven alternative to 6T designs [4-5]. However, their adoption is limited by fundamental constraints, including data-dependent writability arising from single-transistor access and a single bitline structure that relies on single-ended sensing. This sensing mechanism is more susceptible to noise and slower, which limits bitline-length scalability [4-5]. Transmission-gate (TG) access devices eliminate data-dependent write behavior by enabling full-swing signal transfer for both logic levels. However, in nanosheet devices, the additional access device introduces an area overhead, offsetting the density advantage of 5T. This work shows that CFET fundamentally alters this trade-off. The proposed 5T+1 CFET SRAM (**Fig. 2(c)**) exploits vertical n/p stacking to realize a TG-access device within a 5T footprint. To enable differential sensing, a dual-bitline-per-cell-height organization is proposed, where two bitlines within the cell height are selectively connected, with one serving as a reference. The bitcell further integrates a split-V_{DD} write-assist [6] to improve writability. DTCO-based evaluation shows that the proposed bitcell achieves a 12.5% bitcell area reduction relative to 6T, while maintaining RSNM and >150 mV five-percentile (p05) HSNM for unselected cells under split-V_{DD} write operation. In addition, the design achieves a 21% reduction in standby leakage power and a 7–15% faster write.

5T+1 CFET SRAM DTCO: Fig. 3 illustrates the layout of the baseline 6T CFET SRAM with PFET access devices. To realize a six-device bitcell, the unused top-tier nanosheet NFETs above the access devices are selectively removed using the top nanosheet cut module introduced in prior work [1]. **Fig. 4** shows the layout of the proposed 5T+1 CFET SRAM, which extends the 5T design by incorporating a TG-access device while preserving the 5T footprint. Owing to CFET vertical integration, the TG is realized without area penalty. A diffusion break is introduced to isolate the internal storage node from adjacent cells **(Fig. 4)**. Unlike the 6T CFET SRAM, the proposed design retains the top-tier NFET, eliminating the need for the top-nanosheet cut module and enabling more efficient utilization of the CFET stack. For improved writability, the two inverters are supplied by split-V_{DD} rails like in [6], routed on the backside BM2 metal. A split-gate process is employed to independently access the TG transistors; **Fig. 5 (a)** shows the corresponding process flow, simulated using SEMulator3D [7]. The resulting cell occupies 0.0079 µm², a 12.5% area reduction versus the 6T baseline **(Fig. 1)**.

To enable a rectangular, abutment-friendly unit cell, two complementary 5T+1 CFET SRAM cell variants are designed as shown in **Fig. 6(a)**. The variants differ in cross-coupling connectivity and geometry, allowing adjacent cells to share contacts and diffusion breaks. **Fig. 6(b)** shows the abutment of four cells forming a rectangular 4×1 unit cell. Within this 4x1 unit cell, two bitlines are routed within a cell height: cells accessed by two adjacent wordlines connect to one bitline, while cells accessed by the next two wordlines connect to the second bitline **(Fig. 6(b))**. This dual-bitline-per-cell-height organization enables differential sensing, where the bitline of an unselected cell serves as a reference. As the reference bitline depends on the selected wordline, additional multiplexing logic is required to select the appropriate reference input for the sense amplifier as shown in the 5T+1 CFET SRAM array-level organization of **Fig. 7**. This selection is determined by the row address, enabling correct pairing of data and reference bitlines without modifying the sense-amplifier architecture. The column-selective split-V_{DD} control circuitry, shown in **Fig. 8(a)**, temporarily reduces the V_{DD} of one inverter during write to weaken its pull-up device depending on the target data value. Owing to the TG-access device, bitlines are precharged at $0.5{\cdot}V_{DD}$ during standby while preserving the sensing of both logic levels **(Fig. 8(b))**. As shown in **Fig. 8(c)**, this biasing scheme reduces standby leakage power by ~21% compared to 6T, in which bitlines are precharged to 0 V, resulting in higher subthreshold leakage under full-swing biasing.

DTCO Flow and Results: The SRAM DTCO flow starts from the physical layout, from which a full 3D bitcell structure (**Fig. 5(b)**) is constructed using SDE [8], followed by parasitic extraction and netlist generation using Raphael FX [8]. The channel region was modeled with a TCAD-calibrated BSIM-CMG model [9]. The extracted netlist was then used to evaluate both bitcell and array level SRAM KPIs using Spectre [10]. Monte Carlo analysis includes global process variations modeled using worst-case (±3σ) corner vectors and local threshold-voltage mismatch sampled with Gaussian distributions. The proposed 5T+1 CFET SRAM with split-V_{DD} exhibits comparable read static noise margin (RSNM) to the 6T across all process corners **(Fig. 9(a))**. The impact of split-V_{DD} on unselected cells during write is limited. During read and hold, hold static noise margin (HSNM) remains comparable to 6T, while during write, the p05 HSNM of unselected cells sharing the same column V_{DD} remains >150 mV **(Fig. 9(b))** for a 20% V_{DD} reduction. Write-0 and write-1 margin results **(Fig. 10)** show a significant improvement for the 5T+1 CFET SRAM with split-V_{DD} bitcell compared to the bitcell without split-V_{DD}, which exhibits negative write margins for write-0 due to weaker NFET drive relative to the PFET under optimized orientation and stress [9]. **Fig. 11** shows the read delay and write time for increasing array sizes. The proposed 5T+1 CFET SRAM with split-V_{DD} incurs a ~13-30% read-delay penalty, primarily due to reduced bitcell read current under $0.5{\cdot}V_{DD}$ precharge bias, while achieving 7–15% faster write compared to the 6T due to the write assist.

Conclusions: This work introduced the 5T+1 CFET SRAM, a CFET-enabled TG-access bitcell achieving a 5T footprint. It integrated a dual-bitline-per-cell-height organization and split-V_{DD}. DTCO showed 12.5% area and 21% leakage reductions, 7–15% faster write versus 6T (Table I), indicating the proposed bitcell as a HD SRAM alternative for CFET nodes.

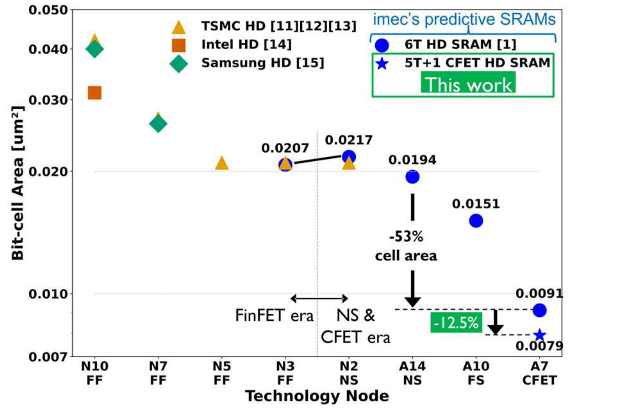

Fig. 1. Bitcell area scaling trends versus technology node, comparing reported HD SRAMs [11-15], in FinFET and NS technologies, with imec's predictive NS and CFET HD SRAM projections [1]. While CFET integration enables significant vertical density scaling, conventional 6T does not fully translate this vertical density benefit into area reduction due to unused stacked devices (Fig. 2(a), Fig. 3). The proposed 5T+1 CFET SRAM (Fig. 2(c), Fig. 4) achieves **a 12.5% bitcell area reduction** at the A7 CFET node compared to 6T.

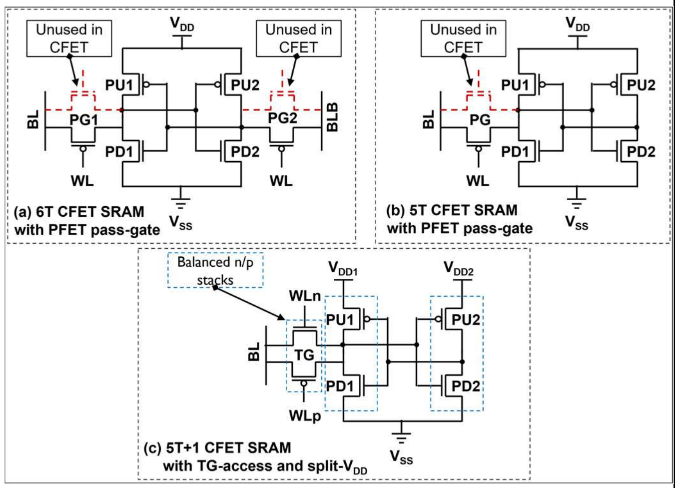

Fig. 2. Schematic comparison of SRAM bitcells in CFET **(a)** 6T CFET SRAM, where the top-tier NFETS above the PGs are unused. **(b)** 5T CFET SRAM, which similarly leaves one device unused. **(c) Proposed 5T+1 CFET SRAM**, where vertical n/p stacking is fully utilized to form a **TG-access device with balanced CFET stacks, enabling efficient exploitation of the CFET architecture**.

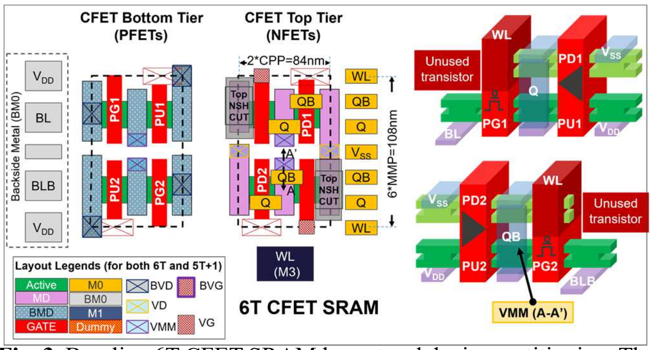

Fig. 3. Baseline 6T CFET SRAM layout and device partitioning. The bottom CFET tier hosts PFET devices (two pull-ups (PU) and two pass-gates (PG)), while the top tier hosts the pull-down (PD) NFETs. To realize a six-device topology, unused top-tier nanosheet NFETs above the access devices are selectively removed using the top nanosheet cut module [1], while preserving the gate, enabling front-side wordline access on M3. Frontside metals are utilized for cross-coupling (M0/M1) and V_{SS} routing (M0/M2), while bitlines and V_{DD} are routed on the backside metals (BM0). Vertical metal merge (VMM) [1] inter-tier connectivity is used to form the inverter outputs Q and QB across the CFET stack, as illustrated in the A-A' cross-section.

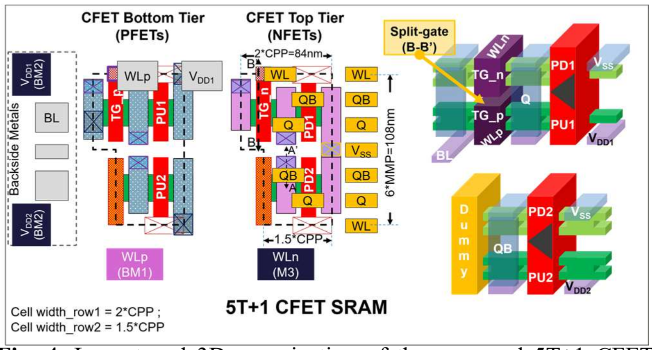

Fig. 4. Layout and 3D organization of the proposed 5T+1 CFET SRAM. The bitcell follows the same CFET organization as the baseline 6T design but retains the top-tier NFET above the access device, **which is vertically stacked with the bottom-tier PFET to form a TG-access device**, eliminating the need for the top nanosheet cut module. A diffusion break is inserted to isolate the internal storage node from adjacent cells. A **split-gate (B-B' cross-section)** is employed to independently access the TG devices, **with wordline routing on the frontside (M3/WLn) and backside (BM1/WLp)**. The two inverters of the bitcell are supplied by independent V_{DD} rails routed on backside metal, enabling column-selective **split-V_{DD} in-bitcell write assist without impacting cell footprint**.

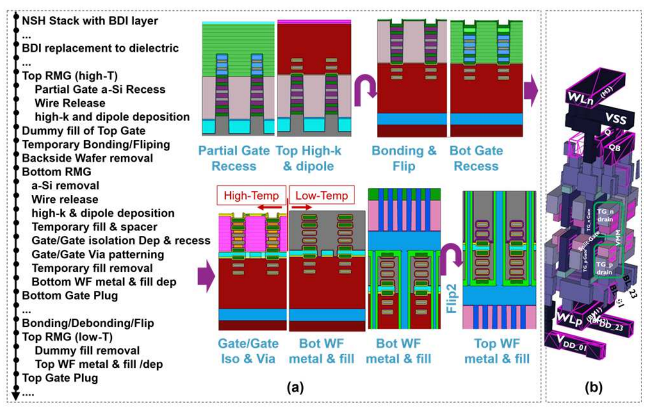

Fig. 5. (a) Monolithic split-gate CFET process flow based on a double-flip approach. Additional bottom-dielectric layers are inserted in the nanosheet stack to protect the channel during gate–gate-via processing. The RMG flow starts with partial a-Si recess for the top device, followed by high-T processing. The wafer is then temporarily bonded and flipped to enable backside access to the bottom gate, after which backside processing is completed. The wafer is flipped back to finish the low-T RMG steps, contacts, and FEOL/BEOL integration. **(b) 3D structure of the proposed bitcell.**

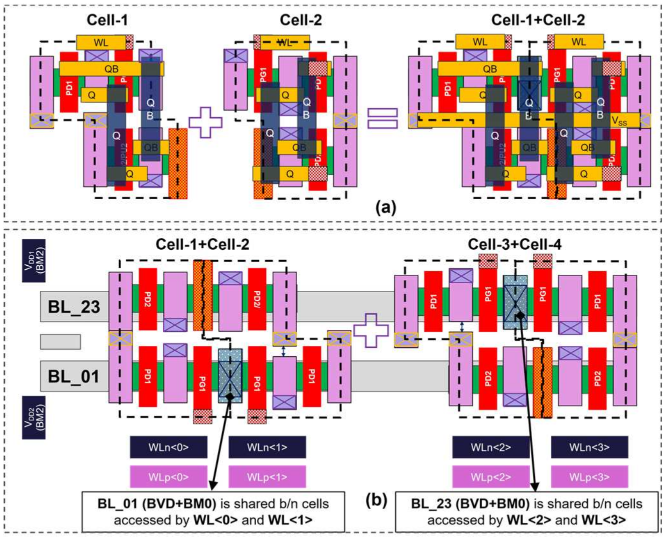

Fig. 6. (a) Two complementary 5T+1 CFET SRAM bitcell variants. **(b)** Four-cell abutment forming a regular 4×1 unit cell, in which two adjacent wordlines share one bitline and the next two share the second, realizing a **dual-bitline-per-cell-height organization**.

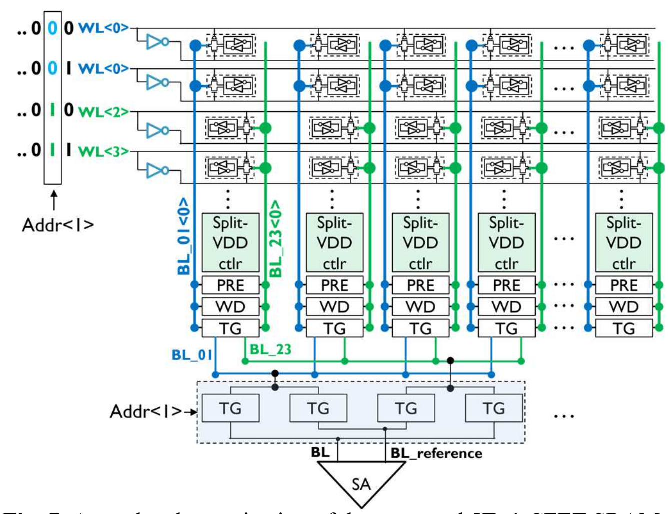

Fig. 7. Array-level organization of the proposed 5T+1 CFET SRAM under a MUX4 configuration, showing **wordline grouping**, **reference-bitline selection logic**, and **column-selective split-V_{DD} control**. The reference bitline is selected based on the **row address Addr<1>** and paired with the data bitline at the sense amplifier.

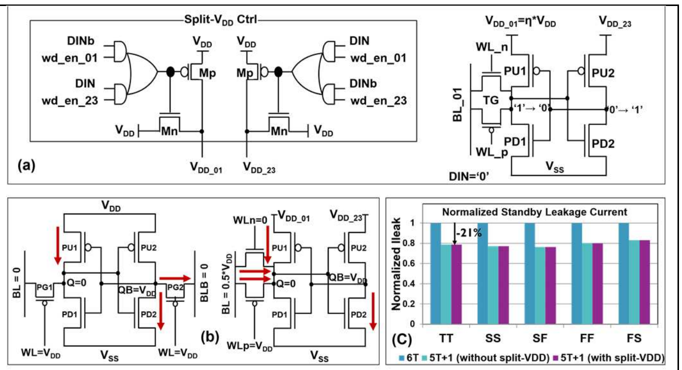

Fig. 8. (a) Column-selective split-V_{DD} control circuitry for the proposed 5T+1 CFET SRAM. During write, the control logic selectively lowers the supply voltage of one inverter (V_{DD_01} or V_{DD_23}) based on the written data and wordline group (wd_en_01/02), temporarily weakening the corresponding pull-up device to improve writability. **(b)** Bitcell biasing under standby operation. **Owing to the TG-access device, both bitlines are biased at 0.5·V_{DD}, preserving the sensing margins while reducing subthreshold leakage compared to full-swing bitline biasing**. **(c)** Normalized standby leakage current across process corners, showing a **~21% leakage reduction** for the 5T+1 CFET SRAM compared to 6T.

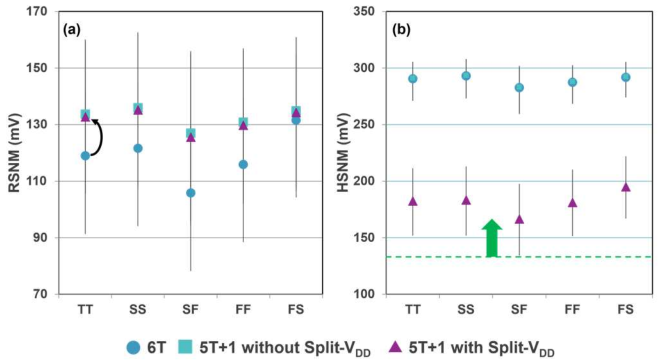

Fig. 9. (a) Read static noise margin (RSNM) and **(b)** hold static noise margin (HSNM) obtained from Monte Carlo simulations across process corners (TT, SS, SF, FF, FS) for 6T, 5T+1 CFET SRAM without split-V_{DD}, and 5T+1 CFET SRAM with split-V_{DD}. **RSNM and HSNM during read and hold operations remain comparable to 6T CFET SRAM**. Under split-VDD operation, **the impact on unselected cells during write** is limited, with the ***p05*** **HSNM remaining above 150 mV across all corners**.

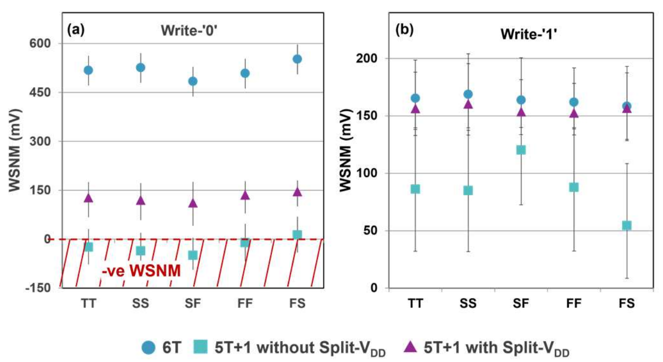

Fig. 10. Monte Carlo write static noise margin (WSNM) for **(a)** write-'0' and **(b)** write-'1' operations across process corners for 6T CFET SRAM, 5T+1 CFET SRAM without split-V_{DD}, and 5T+1 CFET SRAM with split-V_{DD}. The 5T+1 CFET SRAM **without** write assist exhibits **negative write-'0' margin**, indicating poor writability. The proposed bitcell **with split-V_{DD}** write-assist significantly **improves both write-'0' and write-'1' margins**, restoring robust write operation across all corners.

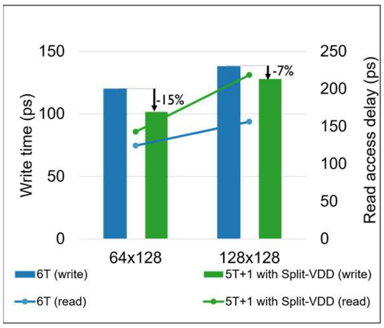

TABLE I Comparison with prior CFET SRAM work [1]

	VLSI'25 [1]	This work
Bitcell	6T CFET	5T+1 CFET
Bitcell area	1x	0.875x
Bitcell Leakage	1x	0.79x
Access device	PFET PG	TG access
Read sensing	Differ-ential (BL/ BLB)	Different-ial (dual-BL/cell-height)
Write assist	-	split-V_{DD}
Split-gate	No	Yes
Top-NS cut	Yes	No
CFET stack utilization	Unbal-anced (4/2)	Balanced n/p stacks

Fig. 11. Array-level write time (bars) and read access delay (lines) for 6T CFET SRAM and 5T+1 CFET SRAM with split-V_{DD}, evaluated for increasing array sizes. The proposed 5T+1 CFET SRAM achieves 7–15% **faster write operation** compared to 6T CFET SRAM, while incurring a ~13-30% **read-delay penalty** due to reduced bitcell current under 0.5·V_{DD} bitline precharge. At the macro level, the reduced bitcell area (-12.5%) enables shorter interconnects and smaller subarrays, **which could partially mitigate the read-delay penalty in large SRAM macros**.

Acknowledgements: This work was partially enabled by the NanoIC pilot line, jointly funded by the Chips Joint Undertaking, through the EU Digital Europe (101183266) and Horizon Europe (101183277) programs, with contributions from participating member states.

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